Math 54 — Week 8

Diagonalization; eigenvectors and linear transformations; complex eigenvalues (Lay 5.3–5.5)

§1 Facts used from Weeks 1–7

(W1) (Week 4) $AP = [\,A\mathbf{p}_1 \;\cdots\; A\mathbf{p}_n\,]$, where $\mathbf{p}_j$ are the columns of P, and $PD = [\,d_1\mathbf{p}_1 \;\cdots\; d_n\mathbf{p}_n\,]$ for $D = \operatorname{diag}(d_1, \ldots, d_n)$. The inverse of a 2×2 matrix with $ad - bc \neq 0$ is

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

(W2) (Weeks 4–6) Invertible matrix theorem: for $n \times n$ P, invertible $\iff$ the columns are independent $\iff$ they form a basis of $\mathbb{R}^n$ $\iff$ $\det P \neq 0$. $\det(AB) = \det A \det B$. Rank theorem: $\operatorname{rank} A + \dim \operatorname{Nul} A = n$.

(W3) (Week 7) λ is an eigenvalue of A $\iff$ $\det(A - \lambda I) = 0$; the eigenspace is $\operatorname{Nul}(A - \lambda I)$. Eigenvectors for distinct eigenvalues are independent. Similar matrices ($B = P^{-1}AP$) have the same characteristic polynomial. The eigenvalues of a triangular matrix are its diagonal entries. $\operatorname{tr} A$ is the sum of the diagonal entries.

(W4) (Week 7) For a basis $\mathcal{B}$ of $\mathbb{R}^n$ and $P_{\mathcal{B}} = [\,\mathbf{b}_1 \cdots \mathbf{b}_n\,]$: $\mathbf{x} = P_{\mathcal{B}}[\,\mathbf{x}\,]_{\mathcal{B}}$ and $[\,\mathbf{x}\,]_{\mathcal{B}} = P_{\mathcal{B}}^{-1}\mathbf{x}$. Taking coordinates is linear and one-to-one, and keeps linear relations.

(W5) (Weeks 4–6) A determinant can be expanded down any column. In a space of dimension d, more than d vectors are dependent, and an independent set can be extended to a basis (Lay 4.5, Thm. 9, Thm. 11).

§2 This week

§2.1 Diagonalization (Lay 5.3)

The question. The difference equation $\mathbf{x}_{k+1} = A\mathbf{x}_k$ has the solution $\mathbf{x}_k = A^k\mathbf{x}_0$. For the matrix below, computing A^{100} by multiplying is hopeless. When can A^k be written down directly for every k?

$$A = \begin{bmatrix} 7 & 2 \\ -4 & 1 \end{bmatrix}$$

Powers of a diagonal matrix are easy: $D = \operatorname{diag}(d_1, \ldots, d_n)$ has $D^k = \operatorname{diag}(d_1^k, \ldots, d_n^k)$, because by (W1) multiplying by a diagonal matrix scales the columns, so the product of two diagonal matrices is diagonal with the products of the diagonal entries. The same is true for any matrix of the form $A = PDP^{-1}$: in the product

$$A^k = (PDP^{-1})(PDP^{-1})\cdots(PDP^{-1})$$

each inner $P^{-1}P$ equals I, so $A^k = PD^kP^{-1}$.

Worked example 1. For the question's matrix:

1. Eigenvalues: $\operatorname{tr} A = 8$ and $\det A = 7 + 8 = 15$, so $\det(A - \lambda I) = \lambda^2 - 8\lambda + 15 = (\lambda - 5)(\lambda - 3)$.

2. Eigenvectors: the first row of $A - 5I$ is $(2, 2)$, so $x_1 = -x_2$ and $\mathbf{p}_1 = (1, -1)$; the first row of $A - 3I$ is $(4, 2)$, so $x_2 = -2x_1$ and $\mathbf{p}_2 = (1, -2)$. Check: $A\mathbf{p}_1 = (5, -5) = 5\mathbf{p}_1$ and $A\mathbf{p}_2 = (3, -6) = 3\mathbf{p}_2$.
3. With these as the columns of P ($\det P = -2 + 1 = -1$),

$$P = \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix}, \quad D = \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix}, \quad P^{-1} = \frac{1}{-1}\begin{bmatrix} -2 & -1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix}.$$

4. By (W1), $AP = [\,A\mathbf{p}_1 \;\; A\mathbf{p}_2\,] = [\,5\mathbf{p}_1 \;\; 3\mathbf{p}_2\,] = PD$, so $A = PDP^{-1}$, and

$$A^k = P\begin{bmatrix} 5^k & 0 \\ 0 & 3^k \end{bmatrix}P^{-1} = \begin{bmatrix} 5^k & 3^k \\ -5^k & -2\cdot 3^k \end{bmatrix}\begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} 2\cdot 5^k - 3^k & 5^k - 3^k \\ 2\cdot 3^k - 2\cdot 5^k & 2\cdot 3^k - 5^k \end{bmatrix}.$$

   For $k = 1$ this gives back A.

The columns of P are eigenvectors of A and the diagonal of D holds their eigenvalues. The next theorem shows that every factorization $A = PDP^{-1}$ has this form.

Definition. A square matrix A is **diagonalizable** if $A = PDP^{-1}$ for some invertible P and diagonal D, that is, if A is similar to a diagonal matrix.

Theorem (The diagonalization theorem (Lay 5.3, Thm. 5))

An $n \times n$ matrix A is diagonalizable if and only if A has n linearly independent eigenvectors. In fact, $A = PDP^{-1}$ with D diagonal if and only if the columns of P are n linearly independent eigenvectors of A; then the diagonal entries of D are the corresponding eigenvalues, in the same order.

The idea

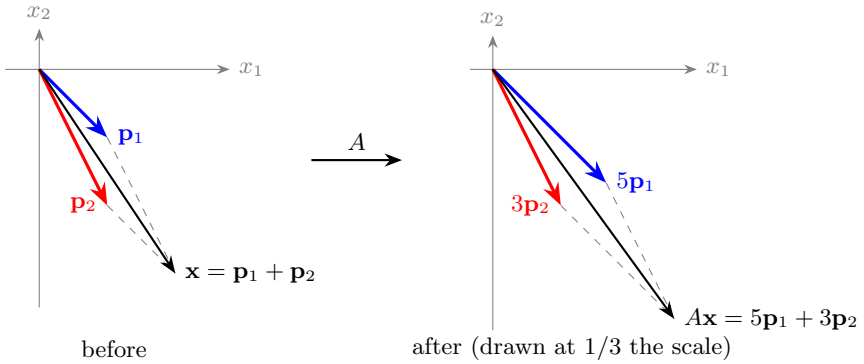
Rewrite $A = PDP^{-1}$ as $AP = PD$ and compare the two sides column by column: column j of AP is $A\mathbf{p}_j$ and column j of PD is $d_j\mathbf{p}_j$. So $AP = PD$ says that each column of P is an eigenvector with eigenvalue d_j, and P being invertible says the columns are independent.

Proof. 1. Let $P = [\,\mathbf{p}_1 \cdots \mathbf{p}_n\,]$ be any $n \times n$ matrix and $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$. By (W1),

$$AP = [\,A\mathbf{p}_1 \;\cdots\; A\mathbf{p}_n\,], \qquad PD = [\,\lambda_1\mathbf{p}_1 \;\cdots\; \lambda_n\mathbf{p}_n\,].$$

2. **Assume $A = PDP^{-1}$.** Multiplying on the right by P gives $AP = PD$, so $A\mathbf{p}_j = \lambda_j\mathbf{p}_j$ for each j. P is invertible, so its columns are independent (W2); in particular none is $\mathbf{0}$. So the $\mathbf{p}_j$ are n independent eigenvectors, with eigenvalues λ_j.
3. **Assume $\mathbf{p}_1, \ldots, \mathbf{p}_n$ are independent eigenvectors**, $A\mathbf{p}_j = \lambda_j\mathbf{p}_j$. By step 1, $AP = PD$. The columns of P are independent, so P is invertible (W2). Multiplying $AP = PD$ on the right by P^{-1} gives $A = PDP^{-1}$.

□



before — after (drawn at 1/3 the scale)

For the matrix of worked example 1, A stretches $\mathbf{p}_1$ by 5 and $\mathbf{p}_2$ by 3. Every $\mathbf{x}$ is a combination of $\mathbf{p}_1$ and $\mathbf{p}_2$, and by linearity A acts on each piece separately; so $A^k\mathbf{x} = 5^k\mathbf{p}_1 + 3^k\mathbf{p}_2$ when $\mathbf{x} = \mathbf{p}_1 + \mathbf{p}_2$.

Diagonalizing a matrix

1. Find the eigenvalues: solve $\det(A - \lambda I) = 0$.
2. For each eigenvalue, find a basis of the eigenspace $\mathrm{Nul}(A - \lambda I)$.
3. If these bases together have n vectors, put them in P as columns and check that P is invertible, so that Theorem 5 applies.
4. Put the eigenvalues in D in the same order as the columns of P. Check $AP = PD$; this avoids computing P^{-1}.

Worked example 2: a repeated eigenvalue that still works.

$$A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix}$$

1. Characteristic polynomial. Subtracting row 3 of $A - \lambda I$ from row 1 (which keeps the determinant) gives the row $(-2 - \lambda,\ 0,\ 2 + \lambda) = (2 + \lambda)(-1, 0, 1)$. Factor out $2 + \lambda$, then add column 1 to column 3 (keeps the determinant) and expand along row 1:

$$\det(A - \lambda I) = (2 + \lambda) \det \begin{bmatrix} -1 & 0 & 0 \\ -3 & -5 - \lambda & -6 \\ 3 & 3 & 4 - \lambda \end{bmatrix} = -(2 + \lambda)\big[(-5 - \lambda)(4 - \lambda) + 18\big]$$
$$= -(2 + \lambda)(\lambda^2 + \lambda - 2) = -(\lambda - 1)(\lambda + 2)^2.$$

   The eigenvalues are 1 and -2, the latter with multiplicity 2.

2. $\lambda = 1$: row reduce $A - I$; the zero row shows the rank is 2.

$$A - I = \begin{bmatrix} 0 & 3 & 3 \\ -3 & -6 & -3 \\ 3 & 3 & 0 \end{bmatrix} \xrightarrow{R_2 + R_3} \begin{bmatrix} 0 & 3 & 3 \\ 0 & -3 & -3 \\ 3 & 3 & 0 \end{bmatrix} \xrightarrow{R_2 + R_1} \begin{bmatrix} 0 & 3 & 3 \\ 0 & 0 & 0 \\ 3 & 3 & 0 \end{bmatrix}$$

   The remaining rows say $x_2 + x_3 = 0$ and $x_1 + x_2 = 0$, so $\mathbf{v}_1 = (1, -1, 1)$.

3. $\lambda = -2$: every row of $A + 2I$ is a multiple of $(1, 1, 1)$, so $\mathrm{Nul}(A + 2I)$ is the plane $x_1 + x_2 + x_3 = 0$, with basis $\mathbf{v}_2 = (-1, 1, 0)$, $\mathbf{v}_3 = (-1, 0, 1)$.
4. Three vectors in all. With

$$P = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}, \qquad D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix},$$

   one checks $AP = PD$ column by column, and $\det P = 1 \neq 0$ confirms that the columns are independent.

Theorem (Distinct eigenvalues (Lay 5.3, Thm. 6))
An $n \times n$ matrix with n distinct eigenvalues is diagonalizable.

Proof. 1. Pick one eigenvector for each of the n eigenvalues.

2. Eigenvectors for distinct eigenvalues are independent (W3), so these are n independent eigenvectors.
3. By Theorem 5, A is diagonalizable.

□

The condition is sufficient, not necessary: the matrix of worked example 2 has two distinct eigenvalues and is diagonalizable, and I has one.

Theorem (Eigenspace dimension and multiplicity (Lay 5.3, Thm. 7a))

If λ_0 is a real eigenvalue of A with algebraic multiplicity m, then $1 \le \dim \operatorname{Nul}(A - \lambda_0 I) \le m$.

The idea

Use a basis of the eigenspace as the first vectors of a new basis of $\mathbb{R}^n$. In that basis the matrix of A has λ_0 on the diagonal and zeros elsewhere in those first columns, so its characteristic polynomial visibly has the factor $(\lambda_0 - \lambda)$ once for each of those columns. Similar matrices have the same characteristic polynomial.

Proof. 1. The eigenspace contains an eigenvector, so its dimension p is at least 1. Let $\mathbf{v}_1, \dots, \mathbf{v}_p$ be a basis of it, and extend it to a basis $\mathbf{v}_1, \dots, \mathbf{v}_n$ of $\mathbb{R}^n$ (W5). Let $P = [\,\mathbf{v}_1 \cdots \mathbf{v}_n\,]$, invertible by (W2), and $B = P^{-1}AP$.

2. For $j \le p$, $P^{-1}\mathbf{v}_j = \mathbf{e}_j$, because $P\mathbf{e}_j = \mathbf{v}_j$. So column j of B is

$$P^{-1}A\mathbf{v}_j = \lambda_0 P^{-1}\mathbf{v}_j = \lambda_0 \mathbf{e}_j,$$

and the first p columns of $B - \lambda I$ each have a single nonzero entry, $\lambda_0 - \lambda$, on the diagonal.

3. Expand $\det(B - \lambda I)$ down column 1, then the resulting minor down its column 1, and so on, p times (W5). Each step contributes a factor $\lambda_0 - \lambda$ with sign $+$, so $\det(B - \lambda I) = (\lambda_0 - \lambda)^p q(\lambda)$ for a polynomial q.

4. B is similar to A, so $\det(A - \lambda I) = \det(B - \lambda I)$ (W3), and $(\lambda - \lambda_0)$ divides it at least p times: $m \ge p$.

□

Theorem (Counting eigenvectors (Lay 5.3, Thm. 7b,c))

Let A be $n \times n$ with distinct real eigenvalues $\lambda_1, \dots, \lambda_r$.

(a) If $\mathcal{B}_k$ is a basis of the eigenspace for λ_k, the vectors of $\mathcal{B}_1, \dots, \mathcal{B}_r$ together are linearly independent.

(b) A is diagonalizable if and only if the dimensions of the eigenspaces add up to n, and this happens if and only if the characteristic polynomial factors completely into real linear factors and the dimension of each eigenspace equals the multiplicity of its eigenvalue.

The idea

(a) applies the distinct-eigenvalue theorem to groups instead of single vectors: in a relation, add up the terms from each eigenspace. Each group sum is an eigenvector or $\mathbf{0}$, and eigenvectors for distinct eigenvalues cannot add up to $\mathbf{0}$. (b) is counting: each eigenspace supplies at most its dimension, and by Theorem 7a at most its multiplicity.

Proof. 1. (a) Take a relation among all the vectors, and let $\mathbf{w}_k$ be the sum of the terms coming from $\mathcal{B}_k$. Then $\mathbf{w}_1 + \cdots + \mathbf{w}_r = \mathbf{0}$, and each $\mathbf{w}_k$ is in the eigenspace for λ_k.

2. If some $\mathbf{w}_k$ were nonzero, the nonzero ones would be eigenvectors for distinct eigenvalues adding up to $\mathbf{0}$, a dependence relation with all weights 1, which (W3) rules out. So every $\mathbf{w}_k = \mathbf{0}$.

3. $\mathbf{w}_k = \mathbf{0}$ is a relation among the vectors of the basis $\mathcal{B}_k$, so all its weights are 0. This holds for every k, so the original relation was trivial.

4. (b) Let d_k be the eigenspace dimensions and m_k the multiplicities. In any independent set of eigenvectors, those with eigenvalue λ_k lie in a d_k-dimensional space, so there are at most d_k of them (W5); so the set has at most $d_1 + \cdots + d_r$ vectors, and by (a) this number is reached. By Theorem 5, A is diagonalizable $\iff d_1 + \cdots + d_r = n$.

5. The characteristic polynomial has degree n, and its real roots λ_k account for $m_1 + \cdots + m_r \le n$ of its factors, with equality exactly when it factors completely into real linear factors. Since $d_k \le m_k$ (Theorem 7a), $d_1 + \cdots + d_r = n$ exactly when $m_1 + \cdots + m_r = n$ and each $d_k = m_k$.

□

Completing the method. If the eigenspace bases of step 2 have fewer than n vectors in total, A is not diagonalizable, by Theorem 7b.

Worked example 3: the same eigenvalues, not diagonalizable.

$$A = \begin{bmatrix} 2 & 4 & 3 \\ -4 & -6 & -3 \\ 3 & 3 & 1 \end{bmatrix}$$

1. Adding row 2 of $A - \lambda I$ to row 1 gives the row $(-2-\lambda,\ -2-\lambda,\ 0) = -(2+\lambda)(1,1,0)$. Factor out $-(2+\lambda)$, subtract column 1 from column 2, and expand along row 1:

$$\det(A - \lambda I) = -(2+\lambda)\det\begin{bmatrix} 1 & 0 & 0 \\ -4 & -2-\lambda & -3 \\ 3 & 0 & 1-\lambda \end{bmatrix} = -(2+\lambda)(-2-\lambda)(1-\lambda) = -(\lambda-1)(\lambda+2)^2,$$

   the same as in example 2.

2. In the matrix below, row 2 is −row 1,

$$A + 2I = \begin{bmatrix} 4 & 4 & 3 \\ -4 & -4 & -3 \\ 3 & 3 & 3 \end{bmatrix},$$

   and rows 1 and 3 are not multiples, so the rank is 2 and by the rank theorem $\dim \mathrm{Nul}(A+2I) = 1$, less than the multiplicity 2.

3. By Theorem 7b, A is not diagonalizable: the eigenspaces have dimensions $1 + 1 = 2 < 3$. Knowing the eigenvalues is not enough; the eigenspace dimensions decide.

Worked example 4: a triangular 4×4 matrix with repeated eigenvalues.

$$A = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 1 & 3 & 4 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}$$

1. A is triangular, so the eigenvalues are $2, 2, 4, 4$ (W3).

2. The two matrices below have rank 2: in $A - 2I$ the nonzero rows are not multiples of each other, and in $A - 4I$ row 3 is $-\frac{1}{2}$(row 1) $-\frac{3}{2}$(row 2), and rows 1, 2 are independent.

$$A - 2I = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 3 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}, \qquad A - 4I = \begin{bmatrix} -2 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 1 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

3. So each eigenspace has dimension $4 - 2 = 2$. For $\lambda = 2$, the equations $x_1 + 3x_2 + 2x_3 = 0$, $x_4 = 0$ give the basis $(-3, 1, 0, 0)$, $(-2, 0, 1, 0)$; for $\lambda = 4$, $x_1 = x_2 = 0$ gives the basis $\mathbf{e}_3$, $\mathbf{e}_4$.

4. $2 + 2 = 4$, so A is diagonalizable (Theorem 7b).

Shortcuts

- **Powers**: $A = PDP^{-1}$ gives $A^k = PD^kP^{-1}$. Reason: the inner $P^{-1}P$ cancel. For a limit as $k \to \infty$: if every eigenvalue is 1 or has $|\lambda| < 1$, then $\lambda^k \to 0$ for the latter, and $A^k \to P\,\mathrm{diag}(\dots)P^{-1}$ with 1's and 0's on the diagonal. Instance: with $D = \mathrm{diag}(1, \frac{1}{2})$, $D^k \to \mathrm{diag}(1, 0)$.
- **Check a diagonalization** with $AP = PD$ (column j: $A\mathbf{p}_j = d_j\mathbf{p}_j$), not by computing P^{-1}. Reason: Theorem 5, step 1. Instance: worked example 1.

- **Quick test**: n distinct eigenvalues imply diagonalizable. A repeated eigenvalue λ of multiplicity m needs $\operatorname{rank}(A-\lambda I) = n-m$, which says its eigenspace has dimension m. Reason: Theorems 6 and 7b and the rank theorem. Instance: worked example 4.
- **Triangular with repeated diagonal entries**: compute the rank of $A-\lambda I$ for each repeated λ. Reason: the quick test. Instance: for the matrix below, $A-2I$ has rank $1 \neq 2-2$, so A is not diagonalizable.

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}, \qquad A - 2I = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

Traps

- The columns of P and the diagonal of D must be in the same order: in worked example 1, $P\,\mathrm{diag}(3,5)P^{-1}$ sends $\mathbf{p}_1 = P\mathbf{e}_1$ to $P\,\mathrm{diag}(3,5)\mathbf{e}_1 = 3\mathbf{p}_1$, while $A\mathbf{p}_1 = 5\mathbf{p}_1$; so it is not A.
- Diagonalizable is not the same as invertible: $\mathrm{diag}(1,0)$ is diagonalizable and singular; the matrix below is invertible and not diagonalizable (eigenvalue 1 twice, $A-I$ of rank 1).

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

- A repeated eigenvalue does not by itself mean "not diagonalizable": $2I$ has the repeated eigenvalue 2 and is diagonal.
- n eigenvalues counted with multiplicity do not give diagonalizability: worked example 3 has three, $1, -2, -2$, and is not diagonalizable.

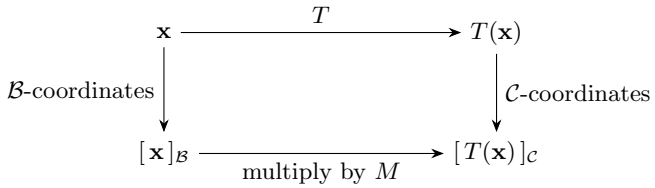
§2.2 Eigenvectors and linear transformations (Lay 5.4)

The question. In worked example 1, $A = PDP^{-1}$ with $D = \mathrm{diag}(5,3)$. What does that say about the transformation $\mathbf{x} \mapsto A\mathbf{x}$ itself, in the basis $\{(1,-1),(1,-2)\}$? And how do we describe a linear transformation on a space like $\mathbb{P}_2$ by a matrix?

The matrix of a linear transformation. Let $T\colon V \to W$ be linear, $\mathcal{B} = \{\mathbf{b}_1, \ldots, \mathbf{b}_n\}$ a basis of V and $\mathcal{C}$ a basis of W. Write $\mathbf{x} = r_1\mathbf{b}_1 + \cdots + r_n\mathbf{b}_n$. By linearity $T(\mathbf{x}) = r_1T(\mathbf{b}_1) + \cdots + r_nT(\mathbf{b}_n)$, and taking $\mathcal{C}$-coordinates is linear (W4), so

$$[\,T(\mathbf{x})\,]_{\mathcal{C}} = r_1[\,T(\mathbf{b}_1)\,]_{\mathcal{C}} + \cdots + r_n[\,T(\mathbf{b}_n)\,]_{\mathcal{C}} = M[\,\mathbf{x}\,]_{\mathcal{B}}, \qquad M = \begin{bmatrix}[\,T(\mathbf{b}_1)\,]_{\mathcal{C}} & \cdots & [\,T(\mathbf{b}_n)\,]_{\mathcal{C}}\end{bmatrix}.$$

M is the **matrix for T relative to $\mathcal{B}$ and $\mathcal{C}$**: its columns are the images of the $\mathcal{B}$ vectors, written in $\mathcal{C}$-coordinates. When $W = V$ and $\mathcal{C} = \mathcal{B}$, M is the **$\mathcal{B}$-matrix** of T, written $[T]_{\mathcal{B}}$, and $[\,T(\mathbf{x})\,]_{\mathcal{B}} = [T]_{\mathcal{B}}[\,\mathbf{x}\,]_{\mathcal{B}}$.



Applying T and then taking $\mathcal{C}$-coordinates (top, then right) gives the same vector as taking $\mathcal{B}$-coordinates and then multiplying by M (left, then bottom).

Worked example: differentiation on $\mathbb{P}_2$. Let $T(a_0+a_1t+a_2t^2) = a_1+2a_2t$ (the derivative), and $\mathcal{B} = \{1, t, t^2\}$.

1. $T(1) = 0$, $T(t) = 1$, $T(t^2) = 2t$, with coordinate vectors $(0,0,0)$, $(1,0,0)$, $(0,2,0)$. These are the columns:

$$[T]_{\mathcal{B}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}.$$

2. To evaluate $T(5+3t-t^2)$, multiply the coordinate vector:

$$[T]_{\mathcal{B}} \begin{bmatrix} 5 \\ 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 3 \\ -2 \\ 0 \end{bmatrix},$$

the coordinate vector of $3-2t$, which is indeed the derivative.

Eigenvectors of a linear transformation. For $T\colon V \to V$, a nonzero $\mathbf{x}$ with $T(\mathbf{x}) = \lambda\mathbf{x}$ is an **eigenvector** of T, and λ an **eigenvalue**. Because coordinates keep linear relations (W4), $T(\mathbf{x}) = \lambda\mathbf{x}$ exactly when $[T]_{\mathcal{B}}[\,\mathbf{x}\,]_{\mathcal{B}} = \lambda[\,\mathbf{x}\,]_{\mathcal{B}}$: the eigenvalues of T are those of $[T]_{\mathcal{B}}$, and the eigenvectors correspond through coordinates. For differentiation on $\mathbb{P}_2$, $[T]_{\mathcal{B}}$ is triangular with diagonal $0,0,0$, so 0 is the only eigenvalue; its eigenspace $\mathrm{Nul}[T]_{\mathcal{B}} = \mathrm{Span}\{\mathbf{e}_1\}$ corresponds to the constant polynomials.

Theorem (Diagonal matrix representation (Lay 5.4, Thm. 8))

Suppose $A = PCP^{-1}$ with P invertible, and let $\mathcal{B}$ be the basis of $\mathbb{R}^n$ formed by the columns of P. Then C is the $\mathcal{B}$-matrix of the transformation $\mathbf{x} \mapsto A\mathbf{x}$. In particular, if $A = PDP^{-1}$ with D diagonal, then D is the $\mathcal{B}$-matrix of $\mathbf{x} \mapsto A\mathbf{x}$.

The idea

In $\mathcal{B}$-coordinates, $\mathbf{x}$ is $P^{-1}\mathbf{x}$ (W4). So the transformation in $\mathcal{B}$-coordinates is: convert back with P, apply A, convert again with P^{-1}; that is, multiplication by $P^{-1}AP = C$. For a diagonal D this says that in the eigenvector basis, the transformation multiplies each coordinate by its eigenvalue.

Proof. 1. By (W4), $\mathbf{x} = P[\,\mathbf{x}\,]_{\mathcal{B}}$ and $[\,A\mathbf{x}\,]_{\mathcal{B}} = P^{-1}(A\mathbf{x})$.

2. So $[\,A\mathbf{x}\,]_{\mathcal{B}} = P^{-1}AP[\,\mathbf{x}\,]_{\mathcal{B}}$ for every $\mathbf{x}$. Since $A = PCP^{-1}$, $P^{-1}AP = C$.

3. The $\mathcal{B}$-matrix is the matrix that turns $[\,\mathbf{x}\,]_{\mathcal{B}}$ into $[\,A\mathbf{x}\,]_{\mathcal{B}}$ for every $\mathbf{x}$; applying both to $[\,\mathbf{b}_j\,]_{\mathcal{B}} = \mathbf{e}_j$ shows that their columns agree. So the $\mathcal{B}$-matrix is C.

□

The question answered. For worked example 1 and $\mathcal{B} = \{(1,-1),(1,-2)\}$, the $\mathcal{B}$-matrix of $\mathbf{x} \mapsto A\mathbf{x}$ is $D = \mathrm{diag}(5,3)$: in $\mathcal{B}$-coordinates, A multiplies the first coordinate by 5 and the second by 3. A linear transformation on $\mathbb{P}_2$ (or any space with a basis $\mathcal{B}$) is described by $[T]_{\mathcal{B}}$, whose columns are the coordinate vectors of the images of the basis vectors.

Worked example: a non-diagonalizable matrix in a good basis.

$$A = \begin{bmatrix} 4 & -9 \\ 4 & -8 \end{bmatrix}$$

1. $\mathrm{tr}\,A = -4$ and $\det A = -32+36 = 4$, so $\det(A-\lambda I) = \lambda^2+4\lambda+4 = (\lambda+2)^2$: the single eigenvalue -2.

2. Both rows of the matrix below are multiples of $(2,-3)$, so it has rank 1,

$$A+2I = \begin{bmatrix} 6 & -9 \\ 4 & -6 \end{bmatrix},$$

so the eigenspace is the line spanned by $(3,2)$, and A is not diagonalizable.

3. Take $\mathcal{B} = \{(3,2),(2,1)\}$, so P has these columns, $\det P = 3 - 4 = -1$, and by (W1) and the 2×2 inverse,

$$AP = \begin{bmatrix} -6 & -1 \\ -4 & 0 \end{bmatrix}, \qquad P^{-1} = \begin{bmatrix} -1 & 2 \\ 2 & -3 \end{bmatrix}, \qquad [\mathbf{x} \mapsto A\mathbf{x}]_{\mathcal{B}} = P^{-1}AP = \begin{bmatrix} -2 & 1 \\ 0 & -2 \end{bmatrix}.$$

4. Not diagonal, but triangular: the first basis vector is an eigenvector, and the second is sent to -2 times itself plus the first.

Facts about similar matrices. Similar matrices are matrices of the same transformation in different bases, which is why they share eigenvalues. If $B = P^{-1}AP$:

- $\det B = \det(P^{-1}) \det A \det P = \det A$ (W2).
- $\operatorname{tr} B = \operatorname{tr} A$, because $\operatorname{tr}(XY) = \operatorname{tr}(YX)$ for square X, Y (both are $\sum_i \sum_k x_{ik} y_{ki}$); take $X = P^{-1}$ and $Y = AP$.
- $B^2 = P^{-1}APP^{-1}AP = P^{-1}A^2P$, so A^2 and B^2 are similar.
- If also $C = Q^{-1}AQ$, then $A = PBP^{-1}$ and $C = Q^{-1}PBP^{-1}Q = (P^{-1}Q)^{-1}B(P^{-1}Q)$, so B and C are similar.
- If $A\mathbf{v} = \lambda\mathbf{v}$, then $B(P^{-1}\mathbf{v}) = P^{-1}APP^{-1}\mathbf{v} = \lambda P^{-1}\mathbf{v}$: the eigenvectors of B are P^{-1} times those of A, not the same vectors.

Shortcuts

- **$\mathcal{B}$-matrix**: column j is $[T(\mathbf{b}_j)]_{\mathcal{B}}$. For $T(\mathbf{x}) = A\mathbf{x}$ on $\mathbb{R}^n$ it is $P^{-1}AP$ with $P = [\,\mathbf{b}_1 \cdots \mathbf{b}_n\,]$. Reason: the derivation above and Theorem 8. Instance: for worked example 1 and the eigenvector basis, $P^{-1}AP = D$.
- **Evaluate T through coordinates**: $[T(\mathbf{x})]_{\mathcal{B}} = [T]_{\mathcal{B}}[\mathbf{x}]_{\mathcal{B}}$. Instance: the derivative of $5 + 3t - t^2$.
- **Eigenvalues of T**: compute those of $[T]_{\mathcal{B}}$. Reason: coordinates keep relations. Instance: differentiation has only the eigenvalue 0.

Traps

- The $\mathcal{B}$-matrix is $P^{-1}AP$, not PAP^{-1}. For worked example 1,

$$PAP^{-1} = \begin{bmatrix} 3 & 0 \\ 6 & 5 \end{bmatrix} \neq D.$$

- Similar matrices need not have the same eigenvectors: $D = P^{-1}AP$ in worked example 1 has the eigenvector $\mathbf{e}_1$ for 5, while $A\mathbf{e}_1 = (7,-4)$ is not a multiple of $\mathbf{e}_1$.

§2.3 Complex eigenvalues (Lay 5.5)

The question. The rotation by 90° below moves every nonzero vector off its line, so it has no real eigenvectors, and $\det(A - \lambda I) = \lambda^2 + 1$ has no real roots. What do the complex roots $\pm i$ mean geometrically?

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

Complex eigenvalues. Allow complex scalars: λ in $\mathbb{C}$ is a (complex) eigenvalue of A if $A\mathbf{x} = \lambda\mathbf{x}$ for some nonzero $\mathbf{x}$ in $\mathbb{C}^n$. The theory of Week 7 carries over unchanged, because its proofs use only addition, multiplication and division by nonzero scalars, which work the same way for complex numbers. In particular the eigenvalues are the complex roots of $\det(A - \lambda I)$, and by the fundamental theorem of algebra (cited) every $n \times n$ matrix has n of them, counted with multiplicity.

Conjugates. For a complex vector or matrix, a bar conjugates every entry, and $\operatorname{Re}\mathbf{x}$, $\operatorname{Im}\mathbf{x}$ are the real vectors with $\mathbf{x} = \operatorname{Re}\mathbf{x} + i \operatorname{Im}\mathbf{x}$. Since conjugation respects sums and products of numbers, $\overline{r\mathbf{x}} = \bar{r}\,\bar{\mathbf{x}}$, $\overline{B\mathbf{x}} = \bar{B}\,\bar{\mathbf{x}}$ and $\overline{BC} = \bar{B}\,\bar{C}$. For a real matrix A, $\bar{A} = A$, so $\overline{A\mathbf{x}} = A\bar{\mathbf{x}}$, and $A\mathbf{x} = A\operatorname{Re}\mathbf{x} + iA\operatorname{Im}\mathbf{x}$ with $A\operatorname{Re}\mathbf{x}$, $A\operatorname{Im}\mathbf{x}$ real: $\operatorname{Re}(A\mathbf{x}) = A\operatorname{Re}\mathbf{x}$ and $\operatorname{Im}(A\mathbf{x}) = A\operatorname{Im}\mathbf{x}$.

Worked example: the rotation by 90°. $\det(A-\lambda I)=\lambda^2+1$, so the eigenvalues are $\pm i$. For $\lambda=i$, the first row of $A-iI$ gives $-ix_1-x_2=0$, so $x_2=-ix_1$ and $\mathbf{x}=(1,-i)$. Check: $A(1,-i)=(i,1)$ and $i(1,-i)=(i,1)$.

- **Conjugate pairs.** If A is real and $A\mathbf{x}=\lambda\mathbf{x}$, conjugating both sides gives $A\bar{\mathbf{x}}=\bar{\lambda}\bar{\mathbf{x}}$. So complex eigenvalues of a real matrix come in pairs $\lambda,\bar{\lambda}$, with conjugate eigenvectors; the second eigenvector need not be computed. Here $\bar{\mathbf{x}}=(1,i)$ is an eigenvector for $-i$.

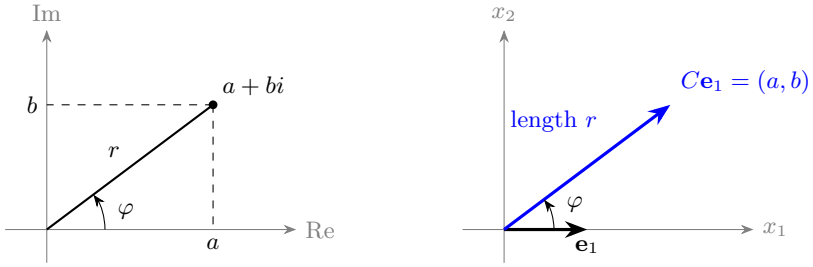
Rotation–scaling matrices. For real a,b, not both 0, let

$$C=\begin{bmatrix}a & -b\\ b & a\end{bmatrix},\qquad \det(C-\lambda I)=(a-\lambda)^2+b^2,\qquad \lambda=a\pm bi.$$

Let $r=|\lambda|=\sqrt{a^2+b^2}$ (also $\sqrt{\det C}$, since $\det C=a^2+b^2$), and let φ be the angle of the point (a,b): $\cos\varphi=a/r$ and $\sin\varphi=b/r$. Then

$$C=r\begin{bmatrix}\cos\varphi & -\sin\varphi\\ \sin\varphi & \cos\varphi\end{bmatrix}:$$

C is the rotation by φ (Week 3) followed by multiplication by r. The modulus of the eigenvalue $a+bi$ is the scale factor r, and its argument is the angle φ.



Left: the eigenvalue $a+bi$ in the complex plane, with modulus r and angle φ. Right: C turns every vector by φ and multiplies its length by r; for example $C\mathbf{e}_1=(a,b)$.

The question answered. The rotation matrix is the rotation–scaling matrix with $a=0$, $b=1$. Its eigenvalues $\pm i$ have modulus $r=1$ (no scaling) and i has argument $\varphi=90°$ (the rotation angle).

Worked example: reading r and φ. For the two matrices below,

$$\begin{bmatrix}\sqrt{3} & -1\\ 1 & \sqrt{3}\end{bmatrix},\qquad \begin{bmatrix}1 & 1\\ -1 & 1\end{bmatrix}.$$

For the first, $a=\sqrt{3}$, $b=1$, $r=2$, and $\varphi=30°$, because $\cos\varphi=a/r=\sqrt{3}/2$ and $\sin\varphi=b/r=1/2$. For the second, $a=1$, $b=-1$, $r=\sqrt{2}$, and $(a,b)=(1,-1)$ is in the fourth quadrant, so $\varphi=-45°$, a clockwise rotation.

Theorem (Real 2×2 matrices with complex eigenvalues (Lay 5.5, Thm. 9))

Let A be a real 2×2 matrix with a complex eigenvalue $\lambda=a-bi$ ($b\neq 0$) and an associated eigenvector $\mathbf{v}$ in $\mathbb{C}^2$. Then $A=PCP^{-1}$, where

$$P=[\operatorname{Re}\mathbf{v}\quad \operatorname{Im}\mathbf{v}],\qquad C=\begin{bmatrix}a & -b\\ b & a\end{bmatrix}.$$

The idea

Write $\mathbf{v}=\mathbf{x}+i\mathbf{y}$ with real $\mathbf{x},\mathbf{y}$. The single complex equation $A\mathbf{v}=\lambda\mathbf{v}$ is two real equations, its real part and its imaginary part. Each says what A does to $\mathbf{x}$ or to $\mathbf{y}$, as a combination of $\mathbf{x}$ and $\mathbf{y}$, and the weights are the entries of C. So $AP=PC$, as in the diagonalization theorem with C in place of D.

Proof. 1. Let $\mathbf{x}=\operatorname{Re}\mathbf{v}$ and $\mathbf{y}=\operatorname{Im}\mathbf{v}$, so $\mathbf{v}=\mathbf{x}+i\mathbf{y}$ and, since A is real, $A\mathbf{v}=A\mathbf{x}+iA\mathbf{y}$ with $A\mathbf{x}$, $A\mathbf{y}$ real.

2. Expand the right side:

$$\lambda\mathbf{v}=(a-bi)(\mathbf{x}+i\mathbf{y})=(a\mathbf{x}+b\mathbf{y})+i(-b\mathbf{x}+a\mathbf{y}).$$

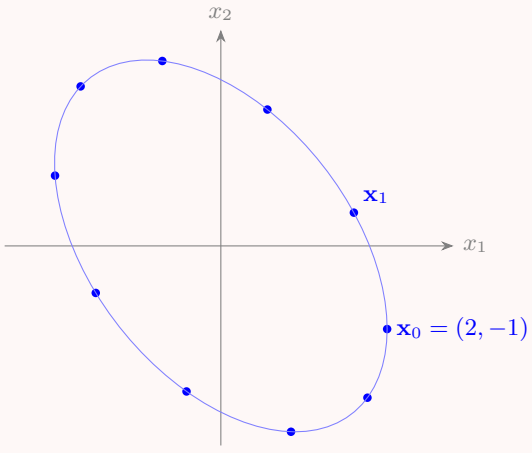
3. The real and imaginary parts of $A\mathbf{v} = \lambda\mathbf{v}$ must agree, so $A\mathbf{x} = a\mathbf{x} + b\mathbf{y}$ and $A\mathbf{y} = -b\mathbf{x} + a\mathbf{y}$.
4. By (W1), these two equations say
$$AP = [\,A\mathbf{x} \;\; A\mathbf{y}\,] = [\,\mathbf{x} \;\; \mathbf{y}\,]\begin{bmatrix} a & -b \\ b & a \end{bmatrix} = PC.$$
5. P is invertible. If $\mathbf{x}, \mathbf{y}$ were dependent, $\mathbf{v}$ would be a complex multiple of a single nonzero real vector $\mathbf{u}$ (one of $\mathbf{x}, \mathbf{y}$), and $\mathbf{u}$ would also be an eigenvector for λ: $A\mathbf{u} = \lambda\mathbf{u}$. The left side is real; the right side is not, because $\mathbf{u}$ is real and nonzero and λ is not real ($b \neq 0$). So $\mathbf{x}, \mathbf{y}$ are independent, and $A = PCP^{-1}$.

□

Worked example (Lay 5.5, Ex. 2).

$$A = \begin{bmatrix} 0.5 & -0.6 \\ 0.75 & 1.1 \end{bmatrix}$$

1. $\operatorname{tr} A = 1.6$ and $\det A = 0.55 + 0.45 = 1$, so $\det(A - \lambda I) = \lambda^2 - 1.6\lambda + 1$, $\lambda = 0.8 \pm 0.6i$, and $|\lambda| = \sqrt{0.64 + 0.36} = 1$.
2. For $\lambda = 0.8 - 0.6i$, the first row of $A - \lambda I$ gives $(-0.3 + 0.6i)x_1 - 0.6x_2 = 0$, so $x_2 = (-0.5 + i)x_1$. With $x_1 = 2$: $\mathbf{v} = (2, -1 + 2i)$.
3. $\operatorname{Re}\mathbf{v} = (2, -1)$, $\operatorname{Im}\mathbf{v} = (0, 2)$, and $a = 0.8$, $b = 0.6$:
$$P = \begin{bmatrix} 2 & 0 \\ -1 & 2 \end{bmatrix}, \qquad C = \begin{bmatrix} 0.8 & -0.6 \\ 0.6 & 0.8 \end{bmatrix}, \qquad A = PCP^{-1}.$$
4. Here $r = 1$, so C is the rotation by the angle with $\cos\varphi = 0.8$, $\sin\varphi = 0.6$, about $36.9°$.
5. The iterates: with $\mathbf{x}_0 = \operatorname{Re}\mathbf{v} = P\mathbf{e}_1$, $\mathbf{x}_k = A^k\mathbf{x}_0 = PC^kP^{-1}P\mathbf{e}_1 = PC^k\mathbf{e}_1$. $C^k\mathbf{e}_1 = (\cos k\varphi, \sin k\varphi)$ lies on the unit circle, and P maps the unit circle onto the curve of points $\cos\theta \operatorname{Re}\mathbf{v} + \sin\theta \operatorname{Im}\mathbf{v}$, an ellipse. If $|\lambda| \neq 1$, then $C^k\mathbf{e}_1 = r^k(\cos k\varphi, \sin k\varphi)$ and the points spiral outward ($r > 1$) or inward ($r < 1$).



The iterates $\mathbf{x}_k = A^k\mathbf{x}_0$ for $\mathbf{x}_0 = \operatorname{Re}\mathbf{v} = (2, -1)$ lie on the ellipse $\{\cos\theta\,(2, -1) + \sin\theta\,(0, 2)\}$; each step advances θ by about $36.9°$.

A 3×3 example. The matrix below has the block C of the previous example in its top left corner. It rotates the x_1x_2-plane by about $36.9°$ and multiplies the x_3-axis by 1.07; its eigenvalues are $0.8 \pm 0.6i$ and 1.07.

$$\begin{bmatrix} 0.8 & -0.6 & 0 \\ 0.6 & 0.8 & 0 \\ 0 & 0 & 1.07 \end{bmatrix}$$

Why $a - bi$ and not $a + bi$. Using the eigenvector $\bar{\mathbf{v}}$ for $\bar{\lambda} = a + bi$ instead replaces $\operatorname{Im}\mathbf{v}$ by $-\operatorname{Im}\mathbf{v}$, which changes P and the sign of b in C. Either choice works if used consistently; Lay's convention pairs $\lambda = a - bi$ with the matrix C above.

Shortcuts

- **Complex eigenvector of a 2×2:** a nonzero row $(p \;\; q)$ of $A - \lambda I$ gives the eigenvector $(q, -p)$ (or any multiple). Reason: $A - \lambda I$ is singular, so the other row is a multiple of this one, and $pq + q(-p) = 0$. Instance: for the rotation by $90°$ and $\lambda = i$, the first row $(-i, -1)$ gives $(-1, i)$, which is -1 times $(1, -i)$.
- **Rotation–scaling data**: from $\lambda = a \pm bi$, the scale factor is $r = \sqrt{a^2 + b^2} = \sqrt{\det A}$, and φ is the angle of the point (a, b), quadrant included. Reason: $\det A = \det C = a^2 + b^2$ (similar matrices). Instance: the reading example.

Traps

- In Theorem 9, P uses $\operatorname{Re}\mathbf{v}$ and $\operatorname{Im}\mathbf{v}$ (real vectors), not $\mathbf{v}$ and $\bar{\mathbf{v}}$, and the eigenvalue used is $a - bi$.
- $\cos\varphi = a/r$ alone does not fix φ: for the second matrix of the reading example, $\cos\varphi = \frac{1}{\sqrt{2}}$ allows $\pm 45°$, and $\sin\varphi = -\frac{1}{\sqrt{2}}$ picks $-45°$.
- Complex eigenvalues of a *real* matrix come in conjugate pairs; a complex matrix need not have that property: $\operatorname{diag}(i, 2)$ has eigenvalues i and 2.

§3 Homework this week

HW7, Sections 5.3–5.4, and HW8, Section 5.5.

Section 5.3.

- **#1, #3** (powers from a given factorization). $A^k = PD^kP^{-1}$.
- **#7** (diagonalize a 2×2). Two distinct eigenvalues, so Theorem 6; one eigenvector each.
- **#13, #19** (diagonalize with a repeated eigenvalue). Find the rank of $A - \lambda I$ for the repeated λ; Theorem 7b (worked examples 2 and 4).
- **#22, #23, #24, #26** (true/false). Theorem 5; worked example 3; $\operatorname{diag}(1, 0)$; I (Theorem 6 is not an "only if").
- **#30** (two eigenvalues, each eigenspace a line, 3×3). $1 + 1 < 3$; Theorem 7b.

Section 5.4.

- **#1, #3, #4** (matrices of transformations). Columns are the coordinate vectors of the images of the basis vectors; for #4(b), compare $T(p)$ with a multiple of p.
- **#5** (evaluate T). $[T(\mathbf{x})]_{\mathcal{B}} = [T]_{\mathcal{B}}[\mathbf{x}]_{\mathcal{B}}$.
- **#7, #8, #9** ($\mathcal{B}$-matrix of $\mathbf{x} \mapsto A\mathbf{x}$). $P^{-1}AP$ with $P = [\,\mathbf{b}_1 \;\; \mathbf{b}_2\,]$; for #9 diagonalize first (Theorem 8).
- **#15, #20** (eigenvectors of a transformation). Check whether $T(p)$ is a multiple of p; a nonzero vector sent to $\mathbf{0}$ is an eigenvector for 0.
- **#22, #23, #27** (similarity facts). The facts about similar matrices in 5.4.

Section 5.5.

- **#3, #5** (complex eigenvalues and eigenvectors). Characteristic polynomial, then one row of $A - \lambda I$ (first shortcut).
- **#10** (read r and φ). The reading example; mind the quadrant.
- **#13, #15** (find P and C). Theorem 9 with $\lambda = a - bi$.
- **#24** (true/false: points on a circle?). The ellipse in Lay's Ex. 2 (last step of that example).

§4 Problem set

P1. Diagonalize the matrix below, and find a formula for A^k.

$$A = \begin{bmatrix} 2 & 3 \\ 4 & 1 \end{bmatrix}$$

P2. Decide whether each matrix is diagonalizable, and diagonalize it if it is.

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}, \qquad B = \begin{bmatrix} 4 & 0 & -2 \\ 2 & 5 & 4 \\ 0 & 0 & 5 \end{bmatrix}$$

P3. A 4×4 matrix has eigenvalues 1, 1, 2, 3, and $\operatorname{rank}(A - I) = 3$. Is A diagonalizable? What if $\operatorname{rank}(A - I) = 2$?

P4. Let $T\colon \mathbb{P}_1 \to \mathbb{P}_2$ be $T(p) = (t+1)p(t)$. Find the matrix of T relative to $\mathcal{B} = \{1, t\}$ and $\mathcal{C} = \{1, t, t^2\}$, and use it to compute $T(3 - 2t)$.

P5. For the matrix below and $\mathcal{B} = \{(1,1), (1,2)\}$, find the $\mathcal{B}$-matrix of $\mathbf{x} \mapsto A\mathbf{x}$. What does it tell you about $(1,1)$ and $(1,2)$?

$$A = \begin{bmatrix} 3 & -1 \\ 2 & 0 \end{bmatrix}$$

P6. For the matrix below, find its complex eigenvalues, an eigenvector for $3 - i$, and P, C with $A = PCP^{-1}$. What are the scale factor and the rotation angle of C?

$$A = \begin{bmatrix} 2 & -1 \\ 2 & 4 \end{bmatrix}$$

P7. Prove: if A is diagonalizable and invertible, then A^{-1} is diagonalizable. If A is diagonalizable, is A^T?

P8. For the matrix below and $\mathbf{x}_0 = (1, 0)$, find $\mathbf{x}_k = A^k\mathbf{x}_0$ and its limit as $k \to \infty$.

$$A = \begin{bmatrix} 3/4 & 1/4 \\ 1/4 & 3/4 \end{bmatrix}$$